Chapter 3
Concurrency Control Techniques

Chapter Outline

- Concepts of Concurrency
- Potential Concurrency Issues
- Concurrency Control
- Locking Techniques
- Concurrency Control based on Timestamp Ordering
- Granularity of Data Items and Multiple Granularity Level Locking

Concepts of Concurrency

- When more than one transaction is being processed against the same database at the same time, the transactions are considered to be concurrent.
- Advantages of concurrency:
  - **increased processor and disk utilization**, leading to better transaction throughput.

  Example: one transaction can be using the CPU while another is reading from or writing to the disk.
  - **reduced waiting time** for transactions: short transactions need not wait behind long ones.

Cont'd…

- If users are only reading data, no data integrity problems will be encountered because no changes will be made in the database.
- In a multiuser environment, simultaneous manipulation of data can result in interference and data loss.
- Solution -> concurrency control.

Concurrency Control

- Concurrency control is important to ensure data integrity when updates occur to the database in a multiuser environment by avoiding any interference between transactions.
- One of the most common concurrency control techniques is **locking** data items.
- A **lock** is a variable associated with a data item that describes the status of the item with respect to possible operations that can be applied to it.

Locking Techniques

- There is one lock for each data item in the database. Thus no two transactions can access the same item concurrently.
- In SQL Server, the basic lock types are:
  - Shared or read (S) lock
  - Exclusive or write (X) lock
- Read operations on an item are not conflicting.
  - allow several transactions to access the same item.
- Must have exclusive lock(X) to write an item.
- **sys.dm_tran_locks** built-in view (for SQL Server).

Locking Problems

- Common locking problems are:
  - **Blocking locks:** Processes are waiting for access to resources.
  - **Deadlock:** A situation in which two processes each hold a lock that the other needs to continue.
    - ✓ Automatically detected and resolved.
    - ✓ Cheapest transaction to rollback is chosen as the victim.

Granularity of Data Items and Multiple Granularity Level Locking

- Size of data items in a database is known as **granularity**.
- Multiple granularity level locking:
  - Allows data items to be of various sizes and define a hierarchy of data granularities.
  - Lock can be requested at any level (database, page, object, key).
  - Additional types of locks, called **intention locks (IS, IX, SIX)**, are needed.

Reading Assignment

- Potential Concurrency Issues
- Concurrency Control based on Timestamp Ordering